

Social Media Sentiment Analysis & Visualization

A social-media post may discuss a service topic and express mixed sentiment. You will prepare text, evaluate a three-class sentiment model, and communicate patterns with counts, proportions and traceable examples. A document is one post; a feature is its numerical representation.

Objective Prepare a 14,640-row CrowdFlower airline corpus for reproducible sentiment classification, aspect exploration and text visualization.

Specific technical objectives

- Preserve all source rows, string identifiers, raw text and the original negative, neutral and positive labels in an audited registry.
- Normalize social-media language while retaining negation, useful hashtag words and an explicit treatment of emoji, mentions and URLs.
- Prevent annotation, author, duplicate-text and vocabulary leakage through independent representatives and a saved training/test manifest.
- Fit and compare a dummy baseline and sparse TF-IDF logistic regression using training-only cross-validation and class-aware evaluation.
- Produce interpretable coefficient, sentiment and aspect visualizations with explicit denominators, limitations and reproducible exports.

Expected learning outcomes

- Distinguish sentiment, topic and aspect, and explain why one post may express several ideas or conflicting opinions.
- Calculate a TF-IDF weight, a three-class probability vector and a cross-entropy contribution by hand.
- Explain how representation choices influence a classifier without mistaking repeated authors or annotation fields for predictive evidence.
- Interpret a confusion matrix, per-class precision/recall/F1 and macro-F1 for an imbalanced corpus.
- Communicate patterns in social text without confusing keyword tags with validated aspects or corpus proportions with public opinion.

Important Guardrails: Sentiment labels describe annotated tone, not verified intent, urgency, satisfaction, toxicity or causation. These historical posts are not a probability sample of travelers. Keep raw text unchanged, redact personal information in displayed examples, and never invent a missing target. Complete this assignment independently; it requires no output from Lab 09B.

Run the demonstration

Use source [A1] if needed; synthetic data cannot meet this assignment. Record source URL, acquisition date, SHA-256 hash and package versions. Companion outputs use `_week09_build/text_nlp_outputs`; student exports use `Lab09A_outputs`. Submit executed work.

Dataset schema and ingestion contract

The February 2015 CrowdFlower file contains 14,640 rows about six US airlines: 9,178 negative, 3,099 neutral and 2,363 positive. Its 14,485 unique `tweet_id` values imply 155

excess duplicate rows; 18 ID groups have conflicting sentiment labels. These are observed source defects. Retain name and tweet_created for author/time audits and airline for descriptive slices. Exclude annotation and engagement fields from predictors. [A1]

Field	Raw type	Canonical meaning	Intentional defect
tweet_id	ID string	Source post identifier	In a stress copy, add whitespace to one ID; detect any collision after trimming. Never round through floating point.
text	string	Original post language	Insert a URL, mention and repeated punctuation in a stress copy; preserve the original and test the normalization rule.
airline_sentiment	string	Negative, neutral or positive annotation	Replace one stress-copy label with an unknown token; flag it rather than guessing its class.
negativereason	nullable text	Post-annotation complaint category	Its value and missingness can reveal sentiment. Demonstrate this leakage trap but exclude both from predictors.
retweet_count	integer text	Engagement at collection	Insert a nonnumeric stress-copy value. Engagement is neither a label nor a permitted predictor.

Injected defects are separate exercises. Preserve 14,640 registry rows with unique source_row keys; tweet_id may repeat. Add text_clean, audit_flags, component_id, model_eligible and split. Quarantine every row in a conflicting-label ID group from supervised modeling, without voting or relabeling. Ineligible and nonrepresentative rows remain visible. Never silently repair unusual source phrases.

Problem 1 Text preparation and leakage diagnosis

Washing a handwritten note should remove a coffee stain without erasing “not”. Text cleaning has the same responsibility. URLs and account handles may add noise or reveal identity, while punctuation, emoji and negation can carry tone. Cleaning is a documented transformation, not a universal deletion recipe.

Tokenization divides text into units; unigrams are single tokens and bigrams preserve adjacent pairs such as “not good”. Lowercasing merges case variants but removes emphasis. Stemming may produce “studi”; lemmatization can produce “study”. Keep a raw column and explain which signals your representation actually captures. Mapping an emoji to a documented token is different from claiming the default word tokenizer understands every emoji. [A2]

Use sublinear term frequency and smoothed inverse document frequency. Here c is a term's count in a document, $df(t)$ counts training documents containing it, N is the training-document count and $\ln$ is the natural logarithm.

$$TF^*(t, d) = 1 + \ln(c)$$
$$IDF(t, D) = \ln \left(\frac{1 + N}{1 + df(t)} \right) + 1$$

The first equation applies when c is positive; set TF^* to zero for absent terms. Multiply TF^* by IDF and normalize each nonzero row to unit Euclidean length. Flag zero rows rather than dividing by zero.

$$x_t = \frac{TF^*(t, d)IDF(t, D)}{\sqrt{\sum_j [TF^*(j, d)IDF(j, D)]^2}}$$

In three training documents, a term appearing once in one document has $TF^* = 1$ and $IDF = \ln(4/2)+1 \approx 1.693$. An unnormalized vector (3,4) becomes (0.6,0.8). Neither calculation uses the test vocabulary.

1.1 Audit missingness, repeated IDs, conflicting-label ID groups, class counts and author/text duplication. Report all denominators and quarantined-row counts. Cross-tabulate negativereason missingness by sentiment to expose annotation leakage. Validate the stress copy separately. Assert conserved rows, unique source_row, string tweet_id and admissible labels. Permit text-derived predictors only; exclude IDs, airline, name, confidence/gold fields, negativereason and retweet_count.

1.2 Show at least five raw/clean pairs covering negation, emoji, hashtags, mentions and URLs. Use deterministic Unicode/whitespace normalization, documented contraction handling, and stable URL/mention markers. Preserve hashtag words and negation. Compare “good” with “not good” before and after an unsafe deletion. Reproduce the two arithmetic examples. Measure token lengths and empty/zero-feature rows, stating what the tokenizer considers a token.

1.3 Connect rows sharing a no missing author or exact nonempty normalized text without using sentiment as a linking feature. Among nonquarantined, valid-time, valid-label, nonempty-text rows, retain the earliest representative per component, breaking ties by tweet_id then source_row. Missing authors do not all form one group. Reserve a stratified 20% as the locked test set with seed 961731; use the remaining 80% for training. Assert disjoint IDs, authors and texts. Key manifests and prediction joins by source_row plus tweet_id. Save the split before feature exploration; fit vocabulary/IDF inside each training fold. This randomized evaluation does not establish future-period performance.

An audit table, labeled class-count bars, raw/clean examples, training token-length distributions and training bigram bars. Full-registry counts check ingestion; detailed feature

decisions use training data. Explain every denominator and distinguish source posts from independent modeling representatives.

Problem 2 Three-class sentiment classification

Everyday bridge: a classifier sorts a stack of messages into three folders. Always choosing the largest folder can appear successful when the stack is imbalanced. A credible model must be compared with that simple strategy and evaluated separately for negative, neutral and positive posts.

For class k , logistic regression forms a score from TF-IDF vector $\mathbf{x}$, intercept b and coefficients β . Softmax converts the three scores to probabilities; prediction uses the largest probability. Keep the class order explicit in every exported matrix. [A3]

$$z_k = b_k + \sum_j \beta_{kj} x_j$$
$$p_k = \frac{e^{z_k}}{\sum_{r=1}^3 e^{z_r}}$$
$$\hat{y} = \arg \max_k p_k$$

Cross-entropy penalizes small probability assigned to the correct label; y_{ik} is one when document i belongs to class k and zero otherwise. The fitted model adds regularization to its training objective. Use library implementations with numerically stable softmax/log-loss calculations.

$$L = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^3 y_{ik} \ln(p_{ik})$$

Arithmetic bridge: scores $(\ln 2, 0, 0)$ produce probabilities $(0.50, 0.25, 0.25)$. If the first class is correct, its loss is $-\ln(0.50) \approx 0.693$; assigning it 0.25 gives approximately 1.386. A probability vector is a model estimate, not evidence of perfect calibration.

2.1 Fit a DummyClassifier using the training class prior and compare its argmax predictions with sparse TF-IDF and L2-regularized multinomial logistic regression. Explain why accuracy alone can hide weak neutral/positive performance. Use unigram/bigram text features, sublinear TF, smoothed IDF and L2 normalization. Declare the dummy strategy and class order. Preserve sparse storage and avoid adding annotation or author metadata to improve apparent accuracy.

2.2 Use three stratified training-only folds with shuffle and seed 961731. Fix max_features at 5,000 and min_df at 2; search C in {0.3, 1, 3}. Keep the vectorizer/classifier in one pipeline. Select mean macro-F1 and retain every candidate's scores, fit time and convergence status. Refit the selected pipeline on all training representatives. Predeclare one bounded comparison, such as unigram-only versus unigram/bigram features, and evaluate it only within the same training folds. Do not search random seeds or inspect test errors during selection.

2.3 Freeze the pipeline and open the locked test set once. Report class support, count and row-normalized confusion matrices, accuracy, macro-F1, per-class precision/recall/F1 and multiclass log loss. For class k , precision divides true positives by predicted positives and recall divides them by actual positives. Declare the zero-denominator convention. Inspect a fixed-seed sample of errors for negation, sarcasm, spelling and mixed sentiment; label these categories as human review, not source annotations. Report results honestly, including weak classes.

$$F1_k = \frac{2P_kR_k}{P_k + R_k}$$
$$\text{MacroF1} = \frac{F1_{\text{negative}} + F1_{\text{neutral}} + F1_{\text{positive}}}{3}$$

For interpretation, plot ten largest positive coefficients per class and show one local contrast between the predicted and runner-up classes. The difference in their scores

equals an intercept difference plus a sum of coefficient-difference contributions. Verify that sum numerically. Correlated words can share or exchange weights; coefficients explain this model's associations, not the causes of a customer's opinion.

Problem 3 Aspect patterns and text visualization

“what people discuss” and “how they feel” are different columns in a listening dashboard. “Baggage” names a subject; “wonderful” expresses tone. A keyword dictionary can help organize posts, but a match is a transparent heuristic that needs manual checking.

Define four keyword tags before reviewing test outcomes: delay, baggage, staff and refund. Record exact phrases and matching boundaries. Allow multiple tags per post and a no-match category. Attach document sentiment to each matched tag for descriptive counts; it is not a validated sentiment label for that aspect.

For aspect a and sentiment class k , report both count n_{ak} and the proportion among posts tagged with a . Multi-label aspects overlap, so summed aspect counts may exceed the number of distinct posts.

$$s_{ak} = \frac{n_{ak}}{n_a}$$

Among 40 posts tagged “baggage”, 24 negative, 10 neutral and 6 positive correspond to 60%, 25% and 15%. Ten posts also tagged “delay” still represent ten posts in the source denominator, not twenty independent observations. An aspect with no matched posts has an undefined proportion.

3.1 Tag and audit Implement the fixed dictionary with vectorized matching and explicit multi-label indicators. Inspect 30 fixed-seed tagged posts plus 20 fixed-seed no-match posts, or all available if fewer. Record false matches, missed paraphrases and ambiguous contexts, retaining source IDs. Report the heuristic's observed audit limitations without

calling it a gold-standard aspect classifier. Keep this descriptive workflow separate from sentiment model selection.

3.2 Produce sentiment count bars, 100% stacked sentiment bars by airline, and an aspect-by-sentiment proportion heatmap with `n_a` displayed. Source-corpus charts use original labels, count rows and disclose duplicates; predicted-label charts name their subset and model. Add a training-only top-bigram chart using document frequency. Use fixed class colors, readable axes, units and a caption stating the population, denominator and whether values are observed or predicted. A word cloud is optional decoration, never the only quantitative evidence.

3.3 Build a one-page findings board containing three defensible observations, their supporting figure and one redacted source example each. Separate the text topic, sentiment label, heuristic tag and model prediction. Identify at least two counterexamples and explain uncertainty from annotation, sampling and normalization. Propose one next data collection or annotation step. Do not retune after test inspection or claim that airline differences establish service quality for all travelers.

Deliverables and submission protocol

- An executed notebook covering Parts 1.1–3.3, calculations, training-only selection, locked-test evaluation, coefficient explanations and the findings board.
- `airline_registry.parquet` with 14,640 rows; `sentiment_split_manifest.csv`, `sentiment_predictions.csv`, `classification_report.csv`, `aspect_sentiment_summary.csv` and `validation_report.json`.
- A data dictionary, source/defect manifest, saved model settings, figures folder and a two-page defense of leakage controls, representation, evaluation and visualization choices.

Submit one ZIP named 961731_studentID_Lab09A.zip. Include the notebook, source script, resolved requirements and source hash. The notebook must run after a kernel restart. Mark every artifact's row population; nonrepresentative registry rows must not be presented as independent test predictions. Predeclare comparisons and keep unfavorable results. Use concise, redacted excerpts rather than exposing account handles or personal details.

Evaluation rubric

Evaluation Rubric 100 points. Credit depends on reproducible evidence, justified decisions and honest interpretation of weak results.

Criterion	Points	Full credit evidence
Diagnosis	25	10: ingestion/leakage audit; 10: text normalization and independent units; 5: correct TF-IDF and probability arithmetic.
Technical Execution	35	15: sparse model and baseline; 10: metrics and coefficient explanations; 10: audited tags and normalized charts.
Methodological Defense	25	10: valid selection/test split; 10: sentiment/aspect distinction and limits; 5: lecture and primary citations.
Automation	15	5: fresh-run reproducibility; 5: passing invariants and exports; 5: complete versioned submission with source and split manifests.

Partial credit: award half for unsupported or irreproducible work; zero for absent evidence, test-driven tuning, fabricated metrics or unresolved execution failures.

Sources and further reading

[A1] CrowdFlower / Figure Eight. Twitter US Airline Sentiment. Retain source attribution and applicable reuse terms; original human labels are the supervised target. <https://www.kaggle.com/datasets/crowdflower/twitter-airline-sentiment>

[A2] Scikit-learn developers. Text feature extraction and common leakage pitfalls. These document the vectorizer and pipeline behavior used here.

[A3] Scikit-learn developers. Logistic regression and classification metrics. Consult the installed version when recording parameter settings.

Local teaching sources: Week09 - NLP for Social Media Analysis.pptx; its Notebook.ipynb/.py companion; Week09 - Notebook Guide.md. Optional model comparisons must preserve the same evaluation contract.

[A4] Lourentzou, Manghnani and Zhai, Adapting Sequence to Sequence models for Text Normalization in Social Media (PDF pp. 1–5, 7–8; supplied arXiv:1904.06100v1).

[A5] Eisner et al., emoji2vec: Learning Emoji Representations from their Description (PDF pp. 2–5; supplied arXiv:1609.08359v2).

[A6] Sarkar, JU_KS@SAIL_CodeMixed-2017: Sentiment Analysis for Indian Code Mixed Social Media Texts (PDF pp. 2–4; supplied arXiv:1802.05737v1).

[A7] Camacho-Collados et al., TweetNLP: Cutting-Edge Natural Language Processing for Social Media (PDF pp. 2–4, 6–8; supplied arXiv:2206.14774v3).

[A8] Aldayel and Magdy, Stance Detection on Social Media: State of the Art and Trends (PDF pp. 5–6; supplied arXiv:2006.03644v5).